METHODOLOGY DESIGN AND INNOVATION PLAN

Project Title

Machine Learning-Based Loan Risk Prediction: A Comparative Study of Ensemble, Oversampling, and Topological Approaches for Credit and Loan Default Assessment

Prepared By

RITHISH R (23CSR178)
NAVEEN SRINIVAS V (23CSR145)
MENAHA R (23CSR130)

Department of Computer Science and Engineering
Academic Year: 2023–2027

Methodology Design

The proposed system is developed to predict the loan default risk of customers using Machine Learning techniques. Initially, the loan dataset is collected from reliable sources and preprocessed by handling missing values, removing duplicate records, and converting the data into a suitable format for analysis. Feature engineering is performed to extract important attributes such as income, credit score, debt ratio, repayment history, employment status, and loan amount. Different Machine Learning models, including XGBoost and CatBoost, are trained and compared to identify the best-performing model. Hyperparameter tuning is applied to optimize the selected model and improve its prediction accuracy and reliability.

After selecting the best model, SHAP and LIME are used to explain the prediction results by identifying the factors that influence each loan decision. Based on the predicted risk level, the system recommends loan eligibility, a suitable loan amount, and an affordable EMI according to the customer's financial profile and repayment capacity. The proposed system also provides a risk score to support better decision-making by financial institutions. Finally, the complete model is integrated into a user-friendly web application that enables real-time loan risk prediction, transparent decision support, and easy access for both customers and bank officers. The system performance is evaluated using metrics such as accuracy, precision, recall, and F1-score.

Planned Innovations

- Compare different Machine Learning models.
- Use SHAP and LIME for explainable predictions.
- Recommend loan eligibility, loan amount, and EMI.
- Improve prediction accuracy through feature engineering.
- Handle imbalanced data for better prediction.
- Develop a real-time web application.
- Provide a customer loan risk score.
- Optimize the model for better performance.